Hybrid Genetic Algorithm and Ant Colony Algorithm for Solving Travelling Salesman Problem

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***Abstract*— The traveling salesman problem (TSP) is a combinatorial optimization problem and a NP-complete problem. It is a well-known problem for comparison of algorithm performance. Many researchers try to solve the TSP by the meta-heuristic algorithms. Ant Colony Optimization (ACO) algorithm is a popular method to solve the TSP. To enhance performance of ACO for solving TSP, the recently proposed improving the ACO algorithm by the results from searching for 2-OPT algorithms that are applied with pheromones of ants. Moreover, when the population of ACO occurs trapping in local optimum, the pheromone of ants is re-initialized to solve trapping in local optimum problem. This algorithm is called Modified Ant Colony Optimization with updating Pheromone by Leader and Re-initialization Pheromone (MACO-LR). However, performance searching of MACO-LR can be improved. In this research, we propose to solve TSP using Hybrid Genetic Algorithm and Ant Colony Algorithm (HGAACO) to optimize the total travelled distance. The proposed technique is tested on twenty-three maps from the Traveling Salesman Problem Library (TSPLIB) and it is composed with MACO-LR. The results from the experiment show the proposed technique can enhance performance searching of MACO-LR.**

Keywords—genetic algorithm, ant colony optimization, 2-OPT algorithm, travelling salesman problem, optimization, re-initialization

I. INTRODUCTION

The traveling salesman problem (TSP) is a well-known optimization challenge and belongs to the NP-complete class of problems. It's frequently used as a benchmark to compare the performance of different algorithms. In 1991, a method called Ant Colony Optimization (ACO) was introduced, inspired by the way ants search for food [1]. ACO has been widely applied to various optimization tasks like job scheduling, network routing, and vehicle routing [2 - 6]. The idea behind ACO is that ants lay down pheromones as they move, with shorter paths having higher pheromone concentrations compared to longer ones [7].

ACO has advantages such as the fast searching for good solutions and strong performance in the tackling TSP [8, 9]. However, a drawback of ACO is that can get stuck in local optimal solutions [10, 11]. To overcome disadvantage of ACO, many researchers proposed many techniques add to ACO process such as 2-OPT [12], 3-OPT [13]. The experiment results of these papers showed that these techniques can get better solutions than the standard ACO.

Jun-man K and Yi Z [11] introduced the ACO-2OPT technique, where the 2-OPT algorithm is applied with ACO at the end of each cycle to enhance solution quality.

Another approach, proposed by Saban Gülcü¸ and et al. [12], involves dividing the colony into multiple colonies and running them in parallel. This method aims to prevent a single colony from getting trapped in a local optimum easily [5, 14]. Each colony searches for solutions independently and exchanges the best solutions with other colonies. If a colony becomes trapped in a local optimum, another colony assists it in escaping. This approach, termed PACO-3OPT, results in better solutions. According to experimental results, the solution quality of PACO-3OPT surpasses that of other compared algorithms.

When using a single colony or ACO-2OPT, ACO is prone to getting stuck in local optima or experiencing premature stagnation in the search process [6]. This algorithm lacks a mechanism to address trapping in local optima. Additionally, while 2-OPT can enhance solutions obtained from ACO searches, it does not aid the search process itself or improve the search conducted by ACO.

If multiple colonies or PACO-3OPT are used, ACO has the capability to address trapping in local optima. When one colony becomes trapped in a local optimum, another colony can assist it in escaping from this situation. However, if both colonies become trapped in close or identical areas, the effectiveness is akin to having a single colony. This scenario is likely because both colonies exchange their best solutions, leading each colony to adjust its paths based on the best solution. Ultimately, both colonies end up being similar, rendering the algorithm ineffective in resolving the issue of trapping in local optima.

Recently, C. Ratanavilisagul and B. Pasaya [15] proposed a novel re-initialization technique apply with ACO when ant colony happens to trap in local optimum. In addition, the 2-OPT applied with ACO. The best results from searching of 2-OPT is taken to improve searching of ACO. The proposed

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technique can get better solutions when it is compared by standard ACO and other compared algorithms. Their technique that proposed is called Modified Ant Colony Optimization with updating Pheromone by Leader and Re-initialization Pheromone or MACO-LR.

Metaheuristics algorithms are optimization methods employed to discover the best solution for difficult problems that cannot be tackled using conventional approaches. Recently, many researchers use hybrid algorithms to solve TSP or similar problems [14, 16, 17]. The experimental results show that using hybrid algorithms can enhance searching performance of algorithms. The hybrid algorithm is using two or more Metaheuristics such as Genetic Algorithm (GA), Ant Colony Algorithm, and Particle Swarm Optimization Algorithm (PSO) concurrently to solve a problem. Each Metaheuristics has the property for differently solving problems.

The experimental results show that the MACO-LR outperforms compared algorithms with regard to the reliability and quality of solutions in all maps. In addition, MACO-LR has high performance for solving TSP. However, performance searching of MACO-LR can be improved and it got better answers from searching. Hence, this research proposed novel hybrid algorithms. The proposed algorithms use GA concurrent to ACO by ACO using a process from MACO-LR. The goal of using hybrid algorithms is to enhance the searching performance of ACO. A set of maps in the TSPLIB [18, 19] is used to compare MACO-LR [15] and the proposed algorithm. The results show that the quality of solutions of the proposed technique is better than other compared algorithms in the TSPLIB.

The remainder of this paper is structured as follows : Section 2 explains the background. Section 3 explains the related works. Section 4 explains the proposed method. Section 5 explains the experiment results. Section 6 and concludes this research and directions for development this research in future work.

II. BACKGROUNDS

A. Travelling Salesman Problem

The traveling salesman problem or TSP [20] is a famous problem in discrete optimization and is known to be difficult to solve. As the number of cities increases, finding the best route becomes increasingly challenging. In this problem, a salesman aims to visit each city exactly once and return to the starting point while covering the shortest possible distance. TSP finds applications in various fields like hardware design and computer networks [21]. A complete weighted graph G = (V, A) can be used to represent a TSP. Where V is the set of n cities and A is the set of paths fully connecting all cities. Each edge (d_{ij}) is the distance between cities i and j. d_{ij} is as follows:

$$d_{i,j} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{1}$$

V = {1,. ., n}, A = {(i, j): i, j ∈ V}, d(i, j) = d(j, i), Edge (i, j) ∈ A

B. Ant colony optimization

This paper demonstrates the use of ACO in tackling the TSP. Here, an ant represents a collection of paths that connect all cities. Therefore, an ant's path serves as a solution to the TSP. The fitness or objective function for the TSP is defined as follows:

$$fitness_of_ant = \sum_{i=1}^{n-1} d_{i,i+1} + d_{n,1} \tag{2}$$

ACO can be described shortly as follow. Initially, each edge has an initial pheromone $\tau_{ij}(0)$ between two cities. The next step is select city of ant. The first city of each ant is randomly selected, and then each ant selects next city according to probability function as follows:

$$P_{i,j}^{k} = \begin{cases} \dfrac{[\tau_{i,j}(t)]^{\alpha}[n_{i,j}]^{\beta}}{\sum_{i \in allowed}[\tau_{i,j}(t)]^{\alpha}[n_{i,j}]^{\beta}} & , j \in allowed \\ 0 & , otherwise \end{cases} \tag{3}$$

Where $\eta_{ij} = \dfrac{1}{d_{ij}}$ is the inverse of the distance, $P_{i,j}^{k}$ is the probability of ant k chooses to move from city i to city j, τ_{ij} is the pheromone, β is a parameter which determines the relative importance of pheromone versus distance (β > 0). The result from formula (3) that causes selecting edges which are shorter and have a greater amount of pheromone. After selection cities of ant has completed, the fitness of ant is calculated by (2). The fitness of each ant is used to update pheromones according to (4). This process continues until a stopping criterion is met.

$$\Delta\tau_{i,j}(t+n) = \rho\tau_{i,j} + \Delta\tau_{i,j} \tag{4}$$

$$\Delta\tau_{i,j} = \sum_{k=1}^{m} \tau_{i,j}^{k} \tag{5}$$

Where ρ is a coefficient such that (1 - ρ) represents the evaporation of trail between time t and t+n, $\tau_{i,j}^{k}$ is the amount of pheromones of the kth ant at its edge between t and t+n, $\Delta\tau_{ij}$ is the sum of new enhanced pheromones at this edge is degree of dissipate for pheromones, $\Delta\tau_{i,j}^{k}$ is calculated by (6), and Q is a constant value, L_k is fitness of ant k.

$$\Delta\tau_{i,j}^{k} = \begin{cases} \dfrac{Q}{L_k} & , if_the_k^{th}_ant_uses_edge(i,j)_in_its_tour \\ 0, & _otherwise \end{cases} \tag{6}$$

C. Genetic Algorithm

Metaheuristics, especially genetic algorithms, have become popular for tackling complex problems, especially when other methods fail to find the best solution. GA is a vital tool in random search, particularly for unclear and complex problems that require significant time and space to find the best solution. They are crucial for finding the optimal solution set for specific designs. GA is efficient for solving highly complex computational problems like the VRP. The process of the genetic algorithm can be summarized. The process starts by creating a random population. Numbers are assigned to each point, representing genes in a chromosome. These numbers encode valid routes based on distances between stations. The population begins with a default route, which is the first chromosome. Then, fitness is calculated, and the best chromosome is identified. The loop continues according to

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defined iterations, refining the population in each iteration with crossover and mutations.

III. Relate Works

A. A Hybrid Genetic and Simulated Annealing Algorithm

M. Sajid, A. Jafar and S. Sharma [14] proposed a hybrid genetic and simulated annealing algorithm (HGSA). HGSA is a hybrid algorithm which combines global optimization methods such as GA and local search SA. The process of HGSA can be summarized as follows. The process of HGSA algorithm starts with a population of randomly generated individuals and then, it generates new population. The chromosome of Capacitated Vehicle Routing Problem (CVRP) is a permutation, meaning each customer in each chromosome appears only once except 0 because 0 is depot. The next step is calculation fitness. The total traveled distance represents the fitness value of the chromosome. The next step is to random number. If the random number is more than 0.7, genetic operators are applied to produce offspring. On the other hand, Simulated Annealing based on swap, inversion, 2-opt, exchange operators are applied to produce offspring. In case of GA, the Nearest-Neighbor crossover operator [15] is applied. The Nearest-Neighbor crossover operator works by making sure that each customer's closest neighbor is visited next. Its goal is to reduce the overall distance traveled by each vehicle. The next step, Selection and Mutation Operators are used in the process. Tournament selection creates new individuals within the group. Mutation operators help maintain variety and avoid early optimal solutions by adding new traits to the group. This is done by randomly adding genes at random spots in existing traits. The swap mutation is employed, where two customers are picked randomly and their positions are exchanged. Then, the local search is applied. In case of SA, The local search applies with simulated annealing algorithm. The local search uses the well-known operators such as swap, inversion, 2-opt, exchange operators. After the process of SA or GA is finished, the next step goes back calculation fitness step and the program terminates after a defined number of iterations.

B. Modified Ant Colony Optimization with updating Pheromone by Leader and Re-initialization Pheromone

In the ACO process, pheromones play a crucial role in guiding ants to form their tours or solutions. When pheromones get stuck in a local optimum, ants tend to follow the same or similar paths as before. Consequently, all ants end up constructing identical solutions repeatedly, making it difficult to find better alternatives [5]. Therefore, altering pheromone levels impacts the search for solutions and helps in escaping local optima.

To address this issue, C. Ratanavilisagul and B. Pasaya [15] suggest resetting a portion of the pheromone when all ants become trapped in a local optimum. This resetting process prompts changes in pheromone levels, encouraging the creation of new tours and enabling the escape from local optima. Typically, resetting should be employed when the entire colony or a significant portion of it becomes trapped or is at risk of getting stuck in a local optimum [22]. This helps in redirecting the ant colony to explore different regions. The easy techniques to indicate the trapping state of the ant colony is to monitor the unchanged number of the best consecutive solution. Thereby, they proposed the re-initialize period (RP) considered from the unchanged number of consecutive solution.

As mentioned earlier, while 2-OPT can improve solutions obtained from ACO, it doesn't improve the overall search process of ACO itself. Updating the pheromone from these improved solutions can lead to even better solutions. These tours that undergo the 2-OPT process are considered good because they have been enhanced. Therefore, it's important to update the pheromone of ACO accordingly to encourage the creation of better solutions. Additionally, ACO should prioritize selecting the best tours to undergo the 2-OPT process. Furthermore, the chosen top-performing ant is referred to as the leader. The outcomes obtained from both ACO and 2-OPT searches can improve the search efficiency simultaneously. From the experimental results show that MACO-LR outperforms compared algorithms with regard to the reliability and quality of solutions in all maps.

IV. Proposed Algorithm

From previous mention, MACO-LR can solve TSP effectively and get a good answer from searching. MACO-LR has techniques that improve answers from searching of ACO and the improved answer is used to increase searching performance of ACO in the next iteration. Hence, searching can get good answers. Moreover, MACO-LR adds the re-initialize process that is used to solve trapping into local optimum. However, the weak point of MACO-LR is the re-initialize process. The re-initialize process can solve or decrease the trapping in the local optimum problem but the re-initialize process of MACO-LR uses the technique of resetting many parts of pheromone. The reset pheromone is similar to starting a new search again. The results from search before reset are abandoned. This method causes waste of resources. In fact, the goal of solving the trapping in the local optimum problem is population search in other areas.

Recently, M. Sajid, A. Jafar and S. Sharma proposed GA that has the process of the Nearest-Neighbor crossover operator. The proposed GA has high performance for solving CVRP. CVRP is a problem that is similar to TSP. So, this research proposed using GA from this research.

The different algorithms cause different searches to behave. The results from different behave of searching affect population search in other areas and solving the trapping in the local optimum problem. So, using an algorithm that can solve that problem effectively is the main algorithm for solving the problem in this case is MACO-LR. Then, using another algorithm that can solve that problem worse than another algorithm is integrated to solve the trapping in the local optimum problem in this case is GA.

From previous mention, to improve searching performance of MACO-LR, this research proposed searching answers by MACO-LR when the trapping in the local optimum problem occurred. It is like the search for answers of ACO is finished. Then, GA is executed to replace the re-initialize process of MACO-LR. In order to GA lead the population to search other areas. GA uses population from ants. So, it is like the search for answers continues from ACO. After the process of GA is

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finished. The best answer from GA is used to update pheromones of ants. The pheromone table is changed by another algorithm. The pheromone table is various. It affects the search in other areas after it is reset. GA helps the population of ACO search other areas and jump out local optimum. Then search using MACO-LR again. This process continues until a stopping criterion is met. The proposed technique is called Hybrid Genetic Algorithm and Ant Colony Algorithm (HGAACO). Pseudo code of HGAACO is shown below:

```
---------- Process of GA ----------
Initial population
For termination generation condition != true do
    If first generation != true
      For each number of population
        Population = offspring
      End For
    End If
    For each calculate fitness of population
      Evaluate fitness of population
    End For
    For each selection of population
      Tournament random selection population from tournament to be parents for create new offspring
    End For
    While termination number of offspring condition != true do
      Crossover of selection 2 population by nearest neighbor to new offspring
      Mutation offspring by swap gene of offspring
      Offspring = new offspring
    End While
    For each number of offspring
      If fitness of each offspring is better than fitness of best population
        Best population = this offspring
      End If
    End For
End For
Calculate best fitness of population
Apply 2-Opt algorithm with best population
Set best population to parameter for update pheromone of ANT
---------- End GA ----------
---------- Process of MACO-LR ----------
Initialize edges (i, j) and pheromone (τij)
For number of update pheromone
    Update Pheromone By GA for First Stage
End For
While termination condition != true do
    Reset Leader
    For each number of ants
      Random the start city
      For until cites in list of an ant is full
        Choose the next city with probability according to formula (3)
        The chose city is not repeat city in list of this ant
        Insert chose city into list of this ant
      End For
      Evaluate fitness of each ant
      If fitness of each ant is better than fitness of solution
        Update fitness of solution = fitness of this ant
      End If
      If fitness of each ant is better than fitness of Leader
        Leader = this ant
      End If
    End For
    Apply 2-Opt algorithm with Leader
    Update pheromone by Leader which is passed 2-Opt according to formula (4)
    Apply Evaporation
    Times of Leader consecutive unchanged ++
    If the times of Leader consecutive unchanged >= re- initialization period (RP)
      For number of update pheromone
        Update Pheromone By GA
      End For
      Set the times of Leader consecutive unchanged to 0
    End If
End While
---------- End MACO-LR ----------
```

V. Experiments and Results

A. Parameters Setting

Parameters are as follows for all experiments: $\alpha = 1$, $Q = 1$, $\rho = 0.05$, $\beta = 1$, $\tau_0 = (n \times L_{nn})^{-1}$ where L_{nn} is the length of list of ant that is produced by the nearest neighbor heuristic and n is the number of cities. The number of experiments of each map is set as 20 runs. The number of ants is set as 100. The non-ACO parameters are as follows: For MACO-LR algorithm, the re-initialization period (RP) = 10, the number of re-initialization (NR) is about 10 percent of the pheromone table size. The parameters for the compared algorithms and the proposed algorithm are configured based on the recommendations outlined in the original paper [15]. The generations of GA is 100 generations per 1 reset. To fairness, the maximum number of iterations of MACO-LR is set as 1500 while the maximum number of iterations of HGAACO is set as 500 because HGAACO has process of GA that is executed so a number of guess answers are more. This research is conducted by a personal computer of AMD FX-8320 with 3.5 GHz CPU and 8 GB RAM and Visual C++ 2010 as the programming language. All maps are used in experiment are from TSPLIB [18, 19].

The measures of algorithm performance in the experiments are as follows: the average best fitness value (ABF) is the average of best fitness in the final iteration from all running (20 runs). ABF indicates the solution searching efficiency of an algorithm. The closer the ABF to the optimum point of a method, the better the method. SR is the success round. SD is the standard deviation. SR and SD indicate the solution searching reliability of an algorithm.

TABLE I .COMPARATIVE RESULTS OF MACO-LR AND HGAACO ON TSP.

Techniques		**MACO-LR**		
Problem name	**Optimum point**	**ABF**	**SD**	**SR**
burma14	3323	3323	0	20
ulysses16	6859	6859	0	20
gr17	2085	2085	0	20
gr21	2707	2707	0	20
ulysses22	7013	7013	0	20
fri26	937	937	0	20
bayg29	1610	1610	0	20
bays29	2020	2020	0	20
Oliver30	420	420	0	20
dantzig42	699	699	0	20
swiss42	1273	1273	0	20
eil51	426	426	0	20
berlin52	7542	7542	0	20
brazil58	25395	25395	0	20
st70	675	675	0	20
pr76	108159	108210	139.126	12
eil76	538	538	0	20
rat99	1211	1211	0	20
eil101	629	629	0	20
lin105	14379	14381	7.14475	13
kroA100	21282	21282	0	20
kroB100	22141	22143	4.96991	12
kroC100	20749	20790.1	51.2967	10

TABLE I .(CONTINUED).

Techniques	**HGAACO**		
Problem name	**ABF**	**SD**	**SR**
burma14	3323	0	20
ulysses16	6859	0	20
gr17	2085	0	20
gr21	2707	0	20
ulysses22	7013	0	20
fri26	937	0	20
bayg29	1610	0	20
bays29	2020	0	20
Oliver30	420	0	20
dantzig42	699	0	20
swiss42	1273	0	20
eil51	426	0	20
berlin52	7542	0	20
brazil58	25395	0	20
st70	675	0	20
pr76	108159	0	20

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eil76	538	0	20
rat99	1211	0	20
eil101	629	0	20
lin105	14379	0	20
kroA100	21282	0	20
kroB100	22141	0	20
kroC100	20749	0	20

The experimental results of Table 1 show that HGAACO can locate the optimum points of all tested maps. The quality solution of HGAACO is better than that of MACO-LR because of its lowest ABF of all tested maps. The reliability of HGAACO is better than that of MACO-LR because of its lowest SD all tested maps. In addition, the SR of MACO-LR is better than that of MACO-LR because of its highest SR of all tested maps. The experimental results show GA that is used in the re-initialize process of MACO-LR can enhance searching performance both quality solution of an algorithm and reliability of an algorithm of MACO-LR. It helps to get better results from searching.

VI. Conclusion

The MACO-LR can solve TSP effectively and get a good answer from searching. Therefore, MACO-LR can solve and decrease trapping in the local optimum problem and improve searching of ACO. However, the weak point of MACO-LR is the re-initialize process. The re-initialize process of MACO-LR is that some parts of pheromone are re-initialized when ant colonies happen to be trapped in the local optimum in order to solve trapping in the local optimum problem. This process can solve or decrease trapping in the local optimum but it is suitable and rarely performance. But, the effect from resetting a large number of ants gives the result is similar to rerun again. The results from searching are rarely performance. This research proposed searching for a hybrid algorithm that uses GA to replace the re-initialize process. It affects the search type that has diversity and can solve or decrease trapping in the local optimum also. The results from GA replace the re-initialize process or Hybrid Algorithm to get better results from searching. The proposed technique is called HGAACO. The experimental results show that the proposed HGAACO outperforms MACO-LR with regard to the reliability and quality of solutions in all maps. In the future, we plan to try to use other Metaheuristics such as PSO, Grey Wolf Optimization, and bee colony algorithm to replace GA. It may get better answers than GA.

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